<Exams Team Use - File Name>

HERIOT WATT UNIVERSITY

School of Mathematical and Computer Sciences

Actuarial Mathematics and Statistics

F78AB

Actuarial and Financial Mathematics

SOLUTIONS

April Examination and Assessment Diet 2024-2025

Duration: 2 Hour(s)

Question 1

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Solution

Payments:	840	720	600	…	0
n: 0	1	2	3	…	8
Alternately, CF:	960	960	960		960
	-120	-240	-360		-960

This can be expressed as $960a_{\overline{8}|}^{0.04} - 120(Ia)_{\overline{8}|}^{0.04}$ (2)

= 6463.44 - 3469.60 (2)

= £ 2,993.84 (1)

Total 5 marks

Question 2

(a) Solution:

Let k be the smallest integer such that: $275\ a_{\overline{k}|}^{0.06} \geq 1000$ (1)

$275\left(\frac{1-(1.06)^{-k}}{0.06}\right) \geq 1000$ (1)

$(1.06)^{-k} \leq 0.7818$

$(1.06)^{k} \geq 1.2791$

$k \geq \frac{\log 1.2791}{\log 1.06}$

$k \geq 4.22$ (1)

$k = 5$ years (1)

Total 4 marks

(b) Solution:

Date	Before Deposit	After Deposit
1/1/21	1,000,000	
31/12/21	1,200,000	1,300,000
31/12/22	1,400,000	1,500,000
31/12/23	1,500,000	1,600,000
31/12/24	1,800,000	

(1)

$(1+i)^4 = (12/10) \times (14/13) \times (15/15) \times (18/16)$ (2)

$(1+i)^4 = 1.453846$

$i = 0.098069 = 9.81\%$ (1)

Total 4 marks

c) Solution:

The decision to not invest is the correct one, because any investment that does not make a positive NPV should be rejected. (1)

His explanation for the rejection is flawed. Assuming that the interest rate used is positive, the NPV of zero means he is not making any return more than the interest rate. The project could be a profitable one, just not profitable enough. (1)

Total 2 marks

Total [10 marks]

Question 3

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(a) Solution:

Capital content = payment – interest paid at time k

$= X - iXa_{\overline{n-(k-1)}|}^{i}$ (1)

$= X(1 - a_{\overline{n-(k-1)}|}^{i})$

$= X(1 - i\left(\frac{1-v^{n-(k-1)}}{i}\right))$ (1)

$= X\left(v^{n-(k-1)}\right)$ (1)

[3 marks]

(b) (i) Solution:

$$100{,}000 = X a_{\overline{25}|}^{0.04}$$
$$= X\left(\frac{1-v^{25}}{0.04}\right)$$
$$= X(15.62208)$$
$$= £6{,}401.20 \quad (1)$$

[1 mark]

(ii) Solution:

$$L_{17} = 6401.20\ a_{\overline{8}|}^{0.04} \quad (1)$$
$$= 6401.20 \times 6.732745 = 43{,}097.65 \quad (1)$$
$$L_{17} = X'\ a_{\overline{8}|}^{0.05} \quad (1)$$
$$43{,}097.65 = X'\ (6.463213)$$
$$X' = 6{,}668.15 \quad (1)$$

Using the equation from part (a) above:

Capital content at time 3 = level payment – interest paid at time 3

$$= X'\left(v^{8-(3-1)}\right) \quad (1)$$
$$= 6{,}668.15\ (1.05)^{-6} = £4{,}975.88 \quad (1)$$

[6 mark]

Total [10 marks]

Question 4

2 marks for initial AV integral
1 for next step
1 for reducing to simple integral
1 for integration,
1 for correct answer
Total 6 marks

$$\text{Accumulated value} = 400\int_0^5 e^{\int_t^5 .035\,ds + \int_5^{15}(.02+.0003s^2)ds}\,dt \quad 2$$

$$= 400\int_0^5 e^{.035(5-t)+\left[.025+\frac{.0003s^2}{3}\right]_5^{15}}\,dt$$

$$= 400\int_0^5 e^{-.035(t-t)+0.3+0.3375-0.1-0.0125}\,dt$$

$$= 400\int_0^5 e^{0.7-0.035t}\,dt$$

$$= 805.501083\int_0^5 e^{-0.035t}\,dt$$

$$= \frac{805.501083}{0.035}\left[1-e^{-0.175}\right]$$

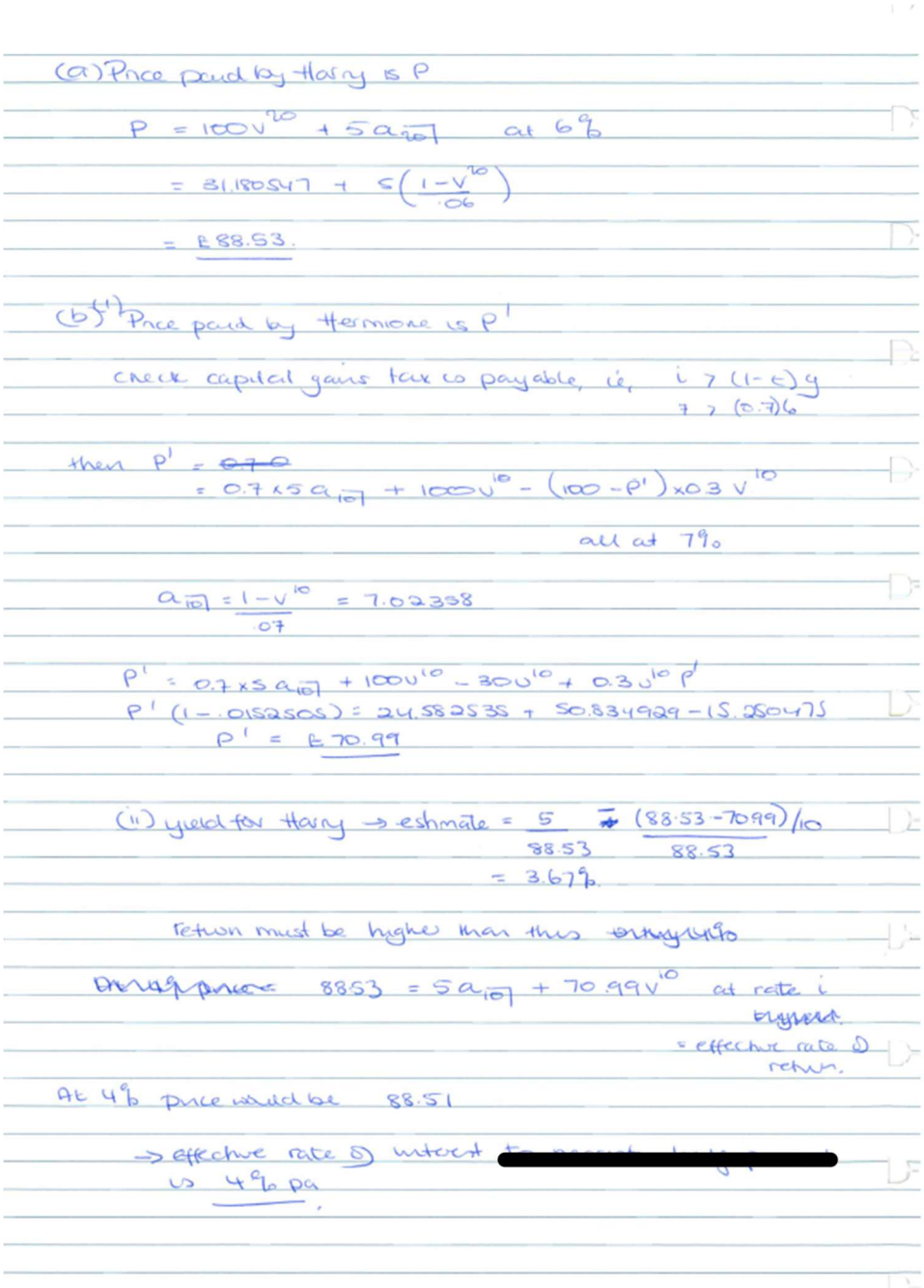
$$= £3694.79.$$

Question 5

a) 1 mark for expressing P in terms of cashflows, 1 for evaluating, 1 for correct answer – **total 3**

b) 1 for checking CGT applies, 2 for expression of P′ 2 separate marks for interim calculations , 1 for correct answer – **total 6**

c) 1 for estimate of yield, 1 for Equation of Value, 1 for check 4% and 1 for final statement – **total 4**



Question 6

Marks as indicated [ ] below**, total 6 marks**

Let 1 year spot-rate = i_1

and 2 year spot-rate = i_2

and $A = \frac{1}{1+i_1}$ $B = \frac{1}{(1+i_2)^2}$

For the 2 year fixed interest stock we have:

$106.50 = \frac{10}{1+i_1} + \frac{10+97}{(1+i_2)^2}$ [1]

$1065 = 10A + 107B$ (1)

From the 2 year par yield we have

$100 = \frac{4.8}{1+i_1} + \frac{100+4.8}{(1+i_2)^2}$ [1]

$100 = 4.8A + 104.8B$ (2)

From (1) $A = \frac{106.50 - 107B}{10}$

From (2) $A = \frac{100 - 104.8B}{4.8}$

$\rightarrow 4.8(106.5 - 107B) = 10(100 - 104.8B)$

$B = 0.91467$ [1]

$A = 0.86302$ [1]

$A = \frac{1}{1+i_1} = 0.86302 \rightarrow i_1 = 0.158716$ [1]

$B = \frac{1}{(1+i_2)^2} = 0.91467 \rightarrow i_2 = 0.045605$ [1]

Question 7

a) 1 for stating conditions, 1 each for PV and DMT assets and liabilities, 1 for concluding statement – **total 6**
b) 1 for equation to determine profit, 1 for correct answer – **total 2**
c) 2 marks for sensible answer

(a) Redingtons first 2 conditions for immunisation

PV(assets) = PV(liabs) [1]

DMT(assets) = DMT(liabs).

$PV(\text{assets}) = 11.657\,v_{6\%}^{2} + 38.813\,v_{6\%}^{30}$

$= 10.374689 + 6.757737$ [1]

$= 17.132$ M.

$PV(\text{liabs}) = 12v_{6\%}^{10} + 25v_{6\%}^{15}$

$= 6.700737 + 10.43163$ [1]

$= 17.132$ M.

$DMT(\text{assets}) = \dfrac{11.657 \times 2v_{6\%}^{2} + 38.813 \times 30v_{6\%}^{30}}{17.132}$

$= \dfrac{20.749377 + 202.732095}{17.132}$

$= \dfrac{223.48147}{17.132}$ [1]

$= 13.0444.$

$DMT(\text{liabs}) = \dfrac{12 \times 10v_{6\%}^{10} + 25 \times 15v_{6\%}^{15}}{17.132}$

$= \dfrac{67.00737 + 156.474398}{17.132}$

$= 13.0444.$ [1]

→ Both conditions are satisfied.

(b) profit/loss = PV(assets) − PV(habs) at new interest rate 7%.

$= 11.657v^2 + 38.813v^{30} - (12v^{10} + 25v^{15})$

$= 10.181675 + 5.09875 - 6.10019 - 9.06115$

$= 0.119$

(c) The spread of the assets is greater than the spread of the liabilities, therefore Redington's 3rd condition is also satisfied, and so we know that profit will occur with small changes in interest rate.

IFoA mapping
Early concepts in 1.1 and 1.2 have been covered in online tests

Q1 *This question covers the topics of Interest Rates, PV, FV, and annuities*.
Syllabus items 1.5.1, 1.5.4, 1.6.1

Q2 *This question covers the topics of yields and discounted cash flows*.
Syllabus 2.3

Q3 This question covers the topics of loans and loan schedules.
Syllabus 2.2.3

Q4 *This question covers the topics of force of interest and continuous cashflows*
Syllabus 1.5.5, 1.6.1

Q5 *This question covers the topics of bonds and tax*
Syllabus 1.7.3, 2.2.2

Q6 *This question covers the topic of term structure of interest rates*
Syllabus 1.7.2

Q7 *This question covers the topics of immunisation*
Syllabus 1.8